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UNIT → 4
INTRODUCTION TO MECHATRONICS

Mechatronics is the integration of mechanical engineering electronics, control System and computer technology to design and develop intelligent systems that can perform tasks automatically, accurately and efficiently.

(#) Mechatronics → Mechanical + Electronics + Control + Computer, working together to make systems smart and automatic.

(#) Main components of Mechatronics →

a) Mechanical Systems → Physical/mechanical parts of the system.

→ Structure → The body of framework that support supports the machine.
→ Design → Arrangement and shape of mechanical parts.
→ Moving parts → Gears, shaft, wheels, arms, belts etc.
→ Actuators → Devices such as motors that produce movement.

ex→ In robots arms.

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b) Electronics → It help the machine sense information and process electrical signals.

→ It includes →

Detect physical quantities such as temperature, speed, light, pressure light or position.

→ Signal conditioning → Makes sensor signals suitable for processing such as amplifying or filtering them.

→ Microcontrollers → Small computers that recieve sensor data and control the system. In automatic washing machine, sensors can detect water level and temperature.

c) Control System → Computers provide the programming and intelligence needed to operate the mechatronic system.

→ feedback loop:→ Compare the target (set point) with the measured output, the difference in the error. The controller adjusts the actuator to reduce it.

→ Common method → PID controller combined proportional, integral and durivative actions. The best controller depends on the machine and its response.

A speed controller increase motor drive when measured falls bellow the target.

④ Computer System:→ Computers provide the programming and intelligence needed to operate the mechatronic system.

→ Typical parts → Embedded software, microprocessor, microcontrollers, memory, communication links and user interface.

→ Real time requirment → Some task finish within a predictable deadline, such as stabilizing a drone or stopping a light at a floor.

A CNC Controller reads a parts program and schedule coordinate axis motion.

System view →

A sensor reports the actual state, the controller compares it will the desired state, an actuator changes the machine and the sensor cheeks the result again.
This reapted is loops what lets many machatronics system correct their own motion.

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(#) Example of Mechatronics →

1. Industrial robot arm →

→ How its work → A rigid arm with several joints is driven by servo motors. Encoders report joint positions, the controller follows a programmed path.

welding, painting, assembly, pallitizing and handing hat or hazardous material

It is high precision and repeatability.

2. CNC Machine →

How its work → A computer reads numerical instruction and coordinate motors that moves a cutting tools and workplace.

uses → moving, turning, drilling and precision engraving.

3. ABS is an Automobile →

→ How its work → wheel speed sensor send measurement to an electric controller. It a wheel is about to look during hard breaking, the hydraulic modulator rapidly adjust breake pressure

→ purpose → Help the driver retain steering control during hard breaking on many road surface.

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4. DSLR Camera →

→ How its works → Autofocus a sensors or image processing software estimate focus error. A small motor moves lens elements until the image reaches the desired focus.

→ Other focus → Expose control, image stabilization and shutter timing also use sensors, electronics and software.

5. Automatic washing machine

→ How it works:→ A controller runs wash, rinse and spin sequence. Sensors may increase water level, temperature, door state, drumspeed

ex→ values, control water, a pump drains it, and a motor turns the drum.

6. Drone (UAV):→

→ How its works → Gyroscopes and accelerometers, measure rotation and motion, other sensors

→ Control idea → The controller repeatdly corrects hit and direction to the aircraft remains stable despite disturbances. such as wind.

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7. Elevator System :→

→ How it works:→ A controller receive floor request and position signals, thth coordinate the drive motor, brakes and doors.

→ Safety and comfort:→ speed ramps, door interlocks, overload sensing and emergency procedures help provids a smooth and safe ride.

8. Smart home system →

→ How its works → Sensor controllers and connected devices co-ordinate lights, heating, securing or appliance.

ex→ A temperature sensor reports room conditions a thermostate switches heating or cooling to keep a chosen range.

History of Mechatronics:→

Mechatronics grew from the need to combine machines with electronic control. As sensor, compact controllers and software improved, equipment could do more than repeat a fixed mechanical motion.

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Word Origin →

→ Yaskawa electric identifier engineer Tetsuro Mori with the term and reports that an application to register it as a trademark was filed in 1969 and registered in A Yaskawa article published in 2000 says the terms remained the company trademark untill 1982, the company later returned it to encourage wider use.

* Timelines →

→ 1969 → Term Introduced

Tetsuro Mori of Yaskawa, coined the word and applied for trademark registration.

→ 1972 → Trademark registered

Yaskawa records the registration in 1972. This corrects the 1971 date shown in the supplied intographic.

→ 1982 → Term released for wider use :→

An industry article reports that the trademark remained registered untill 1982, the word then spread as a general engineering terms.

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→ 1990 → Programmable and digital control grows:

Microprocessors, CNC equipment, sensor and software make machines more flexible and accurate.

→ 2000s - Present - Connected smart system / smart Era.

Robotics, networks, IOT, cloud and edge computing AI and digital twins extend automation across whole factories and services.

⊛ Core Equation →

{Mechanical design structure, motion actuators} + {Electronics Sensors, circuit power} + {Control feedback desision} + {Computing Software data}

=

{Mechatronics Integrated smart system}

Evolution of Mechatronics

* Time line →

a) foundation and Birth (from 1969) →

Tetsuro Mori introduced the name "Mechatronics".

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Described a new field integrating mechanical engineering electronics and control system.

b) Global Recognition (from 1971)

Yaskawa registered official trademark.

c) Growth and global expansion (from 1982)

↳ Trademark renounced / Release to public domain

d) Globus Academic Acceptance (1990's)

Mechatronics include in engineering curriculum.

e) Smart Era / Industry 4.0

Intregation of AI, IOT, Robotics, Big data, cloud.

(#) Stages and Mechatronics :→

* Starting Point (Pre-mechatronics)

a) Pure mechanical system →

characteristics → Mechanical components create motion, operation is mainly mainul setup by a fixed mechanism such as a cam, gear train or linkage.

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Limits → Without electronic sensors and control the machine has little ability to measure changing conditions or correct its own output.

A hand-operated press or a mechanical clock.

→ Phase → 1 Primary Level (1960-1970s)

Concept → Integration

→ Key features → Simple electronics and mechanical units are combined, Relays, switches, timers, and basic actuators automate a limited sequence.

ex→ A washing-machine timer, an automatic door and early electromechanical equipment.

→ Impact → Machines begin to perform repetitive tasks automatically, improving convenience and reducing some routine human effort.

→ Phase → 2 Secondry Level (1980s)

concept → Programming Intelligence

Key features :→ Microprocesors and microcontrollers allows stored programs, more decision making and electronic feedback.

Example → CNC machines follow part programs, electronic braking system use wheel-speed measurements to help regulate breake pressure.

Impact → Higher accuaracy, more flexible operation and less need for constant manual adjustment support industrial automation.

→ Phase -3 Tertiary Level (1990s)

Concept - Software integration and Comportness

Key features → Software become central to product behaviour, better sensor, embedded processors and control algorithms allows compact, responsive system.

ex→ Autofocus cameras, modern industrial robots and coordinated motion, control algorithms allows compact, responsive system equipment.

Impact → Real-time control, improved precision and more capable products.

→ Phase ④ Quaternary Level (2000s - Present) →
concept → (Smart, Connected System)
Key features → Networked sensors and machine, IOT connectivity cloude or edge computing, advanced analytics and in some system, machine learning.

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Example → Smart factories, automotors, vehicles surgical robots and connected building controls

Impact → Machines can share data, co-ordinate tasks and supported predictive maintenance or faster process changes.

(#) Advantage and Disadvantage of Mechatronics →

Advantage →

a) High precision and Accuracy

feature → Sensors and control system reduce human error and provide repeatable, consistent performance.

ex → Industrial welding robots in automobile menufacturing can achieve very high repeatability, the source notes about 0.1 mm accuracy.

b) Automation and Safety →

features → Automated machines can perform repetitive or harardous operations with less direct human exposure.

ex → ABS uses sensors and control logic to help prevent wheel lock during

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c) Flexibility through software →

feature :→ Machine functioning can be changed by simply updating the software (code) without modifying the hardware.

ex→ In CNC machines a new G-code can be uploaded to create a new product design.

d) Compact and Smart Design.

feature :→ Integration of mechanical and electronics components saves space and results in compact and efficient designs.

ex→ In DSLR cameras, the autofocus mechanism is integrated in a very compact size.

e) High productivity and efficiency :→

features → Increases production rate, reduce cycle time and ensures consistent quality with minimum downtime.

ex→ Industrial robotic assembly lines can work 24x7, producing thousands of units with high efficiency.

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7) Improved control and Reliability :→

feature→ Advance sensors, feedback control and intelligent algorithms provides better control, stability and reliable operation.

ex→ Drone technology use sensors and control system to maintain stability and precise movement even in changing conditions.

⑪ Disadvantage :→

a) High Intial cost →

feature→ Intial cost of sensors, microcontrollers, advanted components and designing is very high.

ex→ Converting a traditional assembly line into an automatic robotic line required a huge investement.

b) Complex Maintenance and Repair →

feature→ When a fault accurs, troubleshooting is difficult because both mechanical and electsonic parts have to be checked.

ex→ In modern car, fault diagnosis requires checking both mechanical and electsonics spares.

c) Need for skilled workforce :→

feature → cannot be required (repaired) by ordinary technicians. requires multi disciplinary engineers with knowledge of mechanical, electronics and computer system.

ex→ Repairing CNC machines or 3D printing requires knowledge of computer, mechanical and electronics together.

④ System Dependency and Glitches :→

features → The entire system stops working due to power cut, sensor failure or software bug (glitch)

ex→ If a sensor fails, a fully automatic washing machine stops and shows an error.

⑤ Rapid Technology obsolescence :→

feature → Technology changes very fast, existing system become outdated quickly required upgrade or replacement.

ex→ A 5-year old industrial controller or software may become incompatible with newer system.

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⑥ Integration and Compatibility issues.

features → Components from different manufactures (sensors, controllers, & software) may not be fully compatible, causing integration problem.

ex→ During industrial automation a sensor of one brand may not work properly with a controller of another brand.

→ Application of Mechatronics in various feild

Industrial Robotics: (Automatives and manufacturing)

Overview → Modern assembly lines use mechatronics robots for high-speed and precise manufacturing.

Its used → Car welding
Spray painting
Heavy parts alignment
Material assembly

B) Automated Guided vehicles (AGVs) & (AMRs)

Overview → Driverless vehicles are used to move goods automatically in warehouse and factories

where its used →

Warehouse material handling
factory floor logistics
Hospital (medicine delivery)
Airports and large indoor areas.

© Computer numerical control (CNC) machine

Overview → CNC machine are essential for high-predision and automated manufactoring

where its used → Metal cutting
milling
Turning
Precision component design

(D) Automated conveyor and packaging lines

Overview → Used for high speed packaging, storting and inspection in industries.

where its used →

Beverage bottle filling
food packaging
Pharmaceutical packaging
Defective product rejection.

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(E) Process Control and Continuous production plants →

Overview → Mechatronics helps in monitoring and controlling plant operations for safe and efficient production.

where it is used →

Oil and gases industries
Power plants
Chemical industries
Steel plants.

(F) Agriculture (Smart farming)

Overview → Mechatronics system help in precision farming, automation in aerospace and defense system.

Aircraft control system
Drones and UAVs
Missles guidance systems.

Auto Electronics →

Definition → Auto electronics refers to the electronic components, control units and software systems used in automobiles. these system are designed to improve - vehicle safety, performance, efficiency and comfort.

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① Major Application :→

Defination → ~~Auto~~ (Anti locking braking system) (ABS)

Helps to prevent wheels from locking suddenly during breaking, allowing the driver to maintain better steering control.

Advance driver Assistance System (ADAS) →

Uses sensors, cameras and control electronics for functions such as automatic emergency breaking, lane-departure warning and blind-spot monitoring.

Engine Control Unit (ECU) →

Manages engine-related parameters such as fuel injection and the air-fuel mixture to support performance and mileage.

Electric Vehicle (EV) →

Includes the battery management system (BMS) and motor controllers that monitor the battery and control electric-motor operation.

④ How the electronic control chain work →

Modern vehicle Combine { Sensors → electronic control units → Software decisions → actuators }

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Sensors collects information such as wheel speed, temperature, pressure, vehicle position.

③ Why Auto Electronic Matters :→

Improve safety through faster sensing, monitoring and automated assistance.

Improves performance by controlling engine and motor operation more precisely.

Supports better fuel/energy efficiency by optimizing operation conditions.

Bionics →

Bionics is the technology of understanding the structure and functioning of nature and living biological systems and applying the understanding with electronic and mechanical devices.

* Bionic Limbs / Prosthetics :→

Artificial hands or legs can use muscle signals such as EMG (electromyography) as an input.

An external procussor interprets the signals and commands motors or mechanism in the artificial limb.

The goal is more natural movement, improved control and greater independence.

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(*) Pacemaker →

A pacemaker is a small electronic device associated with heart-rhythm management.

It helps maintain a regular heartbeat when the heart's natural rhythm ~~is~~ needs electronic support.

(*) Coachler Implants →

Cochlear implants are electronics ear implants for people who have lost hearing ability.

The system uses an implant electrode to provide electrical stimulation that can help the user purceive sound.

(#) Avionics →

Avionics means aviation electronics. It refers to electronic, navigation and communication system used in aircraft, spacecraft, missiles and drones.

(#) Autopilot and flight support →

Autopilot → Allows to aircraft maintain selected flight conditions such as speed and altitude without continuous mannul input.

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